

JEE MAIN REVISION MASTER: APPLICATION OF DERIVATIVES 2026 2 QUESTIONS 3.33% WEIGHTAGE ↑99.4%

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 33 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Monotonicity, Maxima & Minima	17 PYQs	Optimization
Very High	Tangents, Normals & Rate of Change	16 PYQs	Tangents & Rates

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Monotonicity, Maxima & Minima	$f'(x) > 0 \implies \text{Strictly Increasing}$, $f'(c) = 0$ & $f''(c) < 0 \implies \text{Local Max}$, $f''(c) > 0 \implies \text{Local Min}$
Tangents, Normals & Rate of Change	$m = \left(\frac{dy}{dx}\right)_{(x_1, y_1)}$, $\text{Tangent: } y - y_1 = m(x - x_1)$, $\text{Normal: } y - y_1 = -\frac{1}{m}(x - x_1)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Monotonicity, Maxima & Minima (Very High Yield — 17 PYQs)

Core Principle: First and second derivative tests for optimization, global extreme values on closed intervals, and Rolle's/Lagrange's Mean Value Theorems.

■ **Caution / JEE Pitfall:** A critical point occurs where $f'(x) = 0$ OR where $f'(x)$ DOES NOT EXIST (e.g. at $x=0$ for $|x|$).

■ **Speed Trick:** LMVT statement: If f is continuous on $[a, b]$ and differentiable on (a, b) , then $\exists c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A YouTube short video is getting viral according to $f(t) = -2t^3 + 3t^2 + 5$. At what time does the video get maximum number of shares? (\$t \dots\$ " [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "The function $f(x) = \tan x - x \dots$ " [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "The value of C in $(0, 2)$ satisfying the mean value theorem for the function $f(x) = x(x-1)^2, x \in [0, 2]$ is equal to..." [Ans: N/A]

3.2 Tangents, Normals & Rate of Change (Very High Yield — 16 PYQs)

Core Principle: Equations of tangent and normal lines, angle of intersection between two curves, and time derivatives as physical rates.

■ **Caution / JEE Pitfall:** Two curves cut orthogonally if and only if the product of their tangent slopes at the intersection point is -1 ($m_1 m_2 = -1$).

■ **Speed Trick:** Length of sub-tangent = $\left| y / (dy/dx) \right|$; length of sub-normal = $|y \cdot (dy/dx)|$.

Representative Source PYQs:

- **[2026 Shift] Q#2 (MCQ):** "In a Mahakumbh, a drone camera is moving along $3y=x^3-3$. When y -coordinate changes 9 times as fast as x -coordinate, it captures..." **[Ans: C]**
- **[2024 Shift] Q#2 (MCQ):** "For the function $f(x)=x^3-6x^2+12x-3$; $x=2$ is..." **[Ans: N/A]**
- **[2024 Shift] Q#4 (MCQ):** "The length of a rectangle is five times the breadth. If the minimum perimeter of the rectangle is 180 cm, then..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A YouTube short video is getting viral according to $f(t)=-2t^3+3t^2+5$. At what time does the video get maximum number of shares? (t is in hours)

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Correct Answer: None

Solution Analysis: The number of shares is given by $f(t)=-2t^3+3t^2+5$. To find when the video gets the maximum number of shares, we differentiate $f(t)$ with respect to t . $f'(t)=(d/dt)(-2t^3+3t^2+5)$ $f'(t)=-6t^2+6t$. For maximum or minimum value, put $f'(t)=0$ $-6t^2+6t=0$ $6t(1-t)=0$ So, $t=0$ or $t=1$. Now check the nature using the second derivative. $f''(t)=-12t+6$. At $t=1$, $f''(1)=-12(1)+6=-6$. Since this is negative, $f(t)$ has a maximum at $t=1$. So, the video gets the maximum number of shares after 1 hour. Correct option: Option A.

Benchmark Problem 2: KCET 2026 (Q#2)

In a Mahakumbh, a drone camera is moving along $3y=x^3-3$. When y -coordinate changes 9 times as fast as x -coordinate, it captures good quality pictures. Then one of the precise positions of the drone at that instant is

- (A) $(-3,-8)$
- (B) $(3,-8)$
- (C) $(3,8)$ **[Correct]**
- (D) $(-3,8)$

Correct Answer: C

Solution Analysis: The path of the drone is given by $3y=x^3-3$. The statement says that the y -coordinate changes 9 times as fast as the x -coordinate. This means $(dy/dx)=9$. Now differentiate the given equation with respect to x : $3y=x^3-3$ $3(dy/dx)=3x^2$ So, $(dy/dx)=x^2$. At the required instant, $x^2=9$ $x=\pm 3$. Now check the points on the curve. For $x=3$: $3y=3^3-3$ $3y=27-3=24$ $y=8$. So one position is $(3,8)$. Hence the correct option is Option C $(3,8)$.